
Empirical IO Chapter 4

Problem Set with Detailed Answers

Demand Function Estimation

This worked set is designed to accompany Chapter 4 of an empirical industrial organization course. The problems move from consumer theory and continuous demand systems to discrete choice models, multinomial logit, mixed logit, nested logit, and the Berry-Levinsohn-Pakes inversion and GMM framework.

Topics covered

- motivations for estimating demand and welfare analysis
- duality, expenditure functions, Hicksian and Marshallian demand
- AIDS share equations and utility-maximization restrictions
- product-space versus characteristics-space demand
- horizontal and vertical differentiation
- multinomial logit, MLE, NLLS, and the IIA problem
- mixed logit, simulation, and quadrature
- nested logit and within-group shares
- unobserved product quality, price endogeneity, inversion, and instruments
- the BLP contraction mapping and GMM logic

Prepared for study and revision

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1 Foundations: Why Demand Estimation Matters

Question 1. Why is demand estimation central in empirical IO and related fields?

Explain why empirical researchers care about estimating demand functions. Your answer should cover at least IO, macroeconomics, and marketing, and it should mention why price elasticity is a particularly important object.

Answer. Demand estimation is central because preferences are the link between observed choices and welfare. In industrial organization, the most important object is often the price elasticity of demand. Once the researcher knows how quantity demanded changes with price, it becomes possible to infer market power, markups, and deadweight loss. This is why demand estimation sits behind merger simulation, pricing counterfactuals, and consumer-surplus calculations.

In macroeconomics, demand systems matter because the expenditure function tells us the minimum spending needed to reach a given utility level. That makes demand estimation relevant for measuring price indices and the cost of living. In marketing, estimated demand is needed for optimal pricing, product design, advertising, and targeting. More broadly, the same logic applies whenever people or firms choose among alternatives: schools, locations, newspaper formats, retail stores, and so on.

A useful way to summarize the motivation is this: supply-side behavior tells us what firms can do, while demand estimation tells us how consumers value the available alternatives. Without demand, we cannot evaluate welfare, substitution patterns, or the profitability of changing prices or product characteristics.

Question 2. What can the econometrician observe and identify in consumer theory?

Why is consumer-demand analysis different from producer-side production-function estimation? What role does revealed preference play?

Answer. A central difference is that utility itself is never directly observed. On the producer side, output is an observable realization of a production process, so there is at least some direct connection between primitives and data. On the consumer side, the primitive object is a utility function, but the econometrician only sees choices made from feasible sets. As a result, demand functions are the main observable implication of consumer theory.

Revealed preference theory bridges this gap. If observed demand is generated by utility maximization, then it must satisfy restrictions implied by rational, monotone, and well-behaved preferences. Those restrictions allow the researcher to recover part of the preference structure from choice data, even though utility levels themselves are not observed. Identification is therefore usually “up to” an admissible representation of preferences rather than recovery of a cardinal utility function.

So the empirical task is not “read utility off the data.” It is instead “ask whether observed

demand could have been generated by optimizing consumers, and if so, recover the components of preference that are behaviorally meaningful.”

Question 3. State the dual objects associated with a utility function.

Suppose a consumer has utility function $u(q)$ and faces price vector p and expenditure x . Define the expenditure function, indirect utility function, Hicksian demand, and Marshallian demand. Explain why empirical work often begins from the expenditure function rather than from direct utility.

Answer. The expenditure function is

$$e(u, p) = \min_q p \cdot q \quad \text{subject to } u(q) \geq u.$$

It gives the minimum spending needed to attain utility level u at prices p .

The indirect utility function is the inverse object,

$$v(p, x) = \max_q u(q) \quad \text{subject to } p \cdot q \leq x.$$

It gives the maximum utility achievable with budget x at prices p .

The Hicksian demand is obtained from the expenditure function by Shepard's lemma:

$$h_i(u, p) = \frac{\partial e(u, p)}{\partial p_i}.$$

This is compensated demand: how much of good i the consumer buys when utility is held fixed.

The Marshallian demand is ordinary demand,

$$d_i(p, x) = h_i(v(p, x), p),$$

which plugs the utility level implied by budget x into Hicksian demand.

Empirical work often starts from $e(u, p)$ because it is easier to derive useful objects from it. The expenditure function immediately yields Hicksian demands, expenditure shares, and the indirect utility function. In many practical models, solving the primal utility-maximization problem in closed form is difficult, while the dual representation is more tractable for estimation and welfare analysis.

Question 4. Derive the expenditure-share equation.

Let $x = e(u, p)$ denote total expenditure. Show that the expenditure share of good i can be written as

$$w_i = \frac{p_i q_i}{x} = \frac{\partial \ln e(u, p)}{\partial \ln p_i}.$$

Interpret this equation.

Answer. Starting from the expenditure function $e(u, p)$, Shepard's lemma gives

$$q_i = \frac{\partial e(u, p)}{\partial p_i}.$$

Multiply both sides by $p_i/e(u, p)$:

$$\frac{p_i q_i}{e(u, p)} = \frac{p_i}{e(u, p)} \frac{\partial e(u, p)}{\partial p_i}.$$

Since total expenditure is $x = e(u, p)$, the left-hand side is the expenditure share of good i ,

$$w_i = \frac{p_i q_i}{x}.$$

The right-hand side is exactly the derivative of log expenditure with respect to log price,

$$\frac{p_i}{e(u, p)} \frac{\partial e(u, p)}{\partial p_i} = \frac{\partial \ln e(u, p)}{\partial \ln p_i}.$$

Hence,

$$w_i = \frac{\partial \ln e(u, p)}{\partial \ln p_i}.$$

This equation is useful because it rewrites demand in terms of observable shares. Instead of modeling quantities directly, we can model expenditure shares as derivatives of a well-behaved expenditure function. This is exactly what makes systems such as AIDS convenient for empirical work.

2 Continuous Choice, Duality, and AIDS

Question 5. How does the Almost Ideal Demand System arise from PIGLOG preferences?

Give the logic that takes the PIGLOG expenditure function to the AIDS share equation

$$w_i = \alpha_i + \sum_j \gamma_{ij} \ln p_j + \beta_i \ln \left(\frac{x}{P} \right).$$

You do not need to reproduce every intermediate algebraic step, but you should explain the main ingredients clearly.

Answer. The starting point is the PIGLOG class,

$$\ln e(u, p) = (1 - u) \ln a(p) + u \ln b(p),$$

where $a(p)$ and $b(p)$ are homogeneous functions of prices. AIDS chooses flexible functional forms for $\ln a(p)$ and $\ln b(p)$ so that the resulting demand system is a first-order approximation to a broad class of demand systems while still obeying the restrictions of utility maximization. Differentiate $\ln e(u, p)$ with respect to $\ln p_i$. By the expenditure-share equation, this derivative equals w_i . The derivative of $\ln a(p)$ produces a constant term and terms linear in $\ln p_j$. The derivative of the difference between $\ln b(p)$ and $\ln a(p)$ produces the coefficient β_i multiplying a utility-dependent term. After substituting the indirect utility function $u = v(p, x)$ into the share equation, that utility term becomes $\ln(x/P)$, where P is the price index implied by the model.

The result is

$$w_i = \alpha_i + \sum_j \gamma_{ij} \ln p_j + \beta_i \ln \left(\frac{x}{P} \right),$$

with

$$\ln P = \alpha_0 + \sum_k \alpha_k \ln p_k + \frac{1}{2} \sum_k \sum_j \gamma_{kj} \ln p_k \ln p_j.$$

Thus AIDS turns the dual consumer problem into a system of share equations that is flexible, interpretable, and estimable.

Question 6. What do the AIDS restrictions mean economically?

State the adding-up, homogeneity, and symmetry restrictions in the AIDS model. Then explain the economic content of each restriction.

Answer. In AIDS, the main restrictions are:

$$\sum_i \alpha_i = 1, \quad \sum_i \gamma_{ik} = 0, \quad \sum_i \beta_i = 0,$$

for adding up,

$$\sum_j \gamma_{ij} = 0$$

for homogeneity, and

$$\gamma_{ij} = \gamma_{ji}$$

for symmetry.

Adding up. Expenditure shares must sum to one, because all expenditure is allocated across

goods. The coefficient restrictions guarantee that if we add the share equations across goods, the result is exactly one rather than some arbitrary function of prices and expenditure.

Homogeneity. Demand should not change if all prices and total expenditure are multiplied by the same positive scalar. Only relative prices and real expenditure matter. The homogeneity restriction makes the demand system invariant to a pure change in units.

Symmetry. Symmetry comes from integrability and utility maximization. It implies that compensated cross-price effects are reciprocal: the way the expenditure share of good i responds to the price of good j mirrors the response of good j to the price of good i , once everything is expressed within the dual system.

Together these restrictions enforce consistency with consumer optimization. They are not mere normalizations; they are testable implications of the underlying theory.

Question 7. Why is the Stone price index often used, and what are its limits?

The AIDS model uses a nonlinear price index P . Explain why researchers often replace it with the Stone index, and what is gained or lost by doing so.

Answer. The exact AIDS price index is

$$\ln P = \alpha_0 + \sum_k \alpha_k \ln p_k + \frac{1}{2} \sum_k \sum_j \gamma_{kj} \ln p_k \ln p_j.$$

This object is nonlinear in the parameters and in prices. In practice, that can make estimation cumbersome, especially in older computational settings or when the researcher wants a simpler estimating equation.

The Stone index replaces $\ln P$ by a share-weighted price index,

$$\ln P^{\text{Stone}} = \sum_j w_j \ln p_j,$$

where the weights are observed expenditure shares. The gain is computational simplicity: the demand system becomes much easier to estimate and interpret.

The loss is that the Stone index is an approximation, not the exact model-implied price index. That means the resulting estimates need not inherit all the structural exactness of AIDS. In applications where the approximation is good, this may be acceptable. But when accurate welfare or substitution calculations are essential, researchers often prefer the exact or iterated price index.

Question 8. Why does aggregate-demand estimation often require instruments?

Contrast the endogeneity problem in aggregate data with the case of household-level data. Why might aggregate data require as many instruments as the dimension of the price vector?

Answer. With aggregate data, the error term in a demand equation often contains market-level shocks to taste, quality, advertising, and availability. These shocks are precisely the things that firms may observe when setting prices. As a result, prices are typically correlated with the error term, so ordinary least squares is inconsistent.

In a system of demand equations, there is one endogenous price component for each product or price dimension being treated as choice-relevant. To identify the demand system, the researcher therefore needs excluded variables that shift prices but do not directly shift demand. In other words, a system with many endogenous prices usually needs a matching set of instruments.

At the household level, if consumers in the same market face the same competitive price vector, then household-specific taste shocks may be orthogonal to prices after controlling for aggregate market conditions. In that narrower setting, price endogeneity may be less severe. But this does not mean household data magically solve the problem: if prices reflect market power, quality sorting, or endogenous product positioning, instruments are still needed.

3 From Product Space to Characteristics Space

Question 9. Why move from a product-space approach to a characteristics-space approach?

Explain the main advantage of defining utility over product characteristics instead of directly over named products.

Answer. In a product-space approach, each product is its own object in utility, so a market with J products can generate a parameter space that scales badly with J . This becomes cumbersome when there are many products or when the researcher wants to evaluate products that were not observed in the estimation sample.

The characteristics-space approach solves this by assuming consumers value attributes rather than product labels. Goods are inputs into characteristics. Then the dimension that matters is the number of characteristics, not the number of products. This gives two major advantages. First, the model becomes more parsimonious. Instead of estimating a free utility index for every product pair, the researcher estimates how consumers value horsepower, sugar content, screen size, brand, distance, or some other attribute vector. Second, new products can be evaluated naturally. A new product enters the model through its characteristics, so its demand implications can be studied without needing to pre-assign it to an arbitrary “similar product” category.

The trade-off is that the researcher must make stronger maintained assumptions about what

characteristics matter and how they enter utility.

Question 10. How does weak separability shape income effects in discrete choice?

Compare the following two specifications for choice-specific indirect utility:

$$u^*(x_j, z_i) + \alpha(y_i - p_j), \quad u^*(x_j, z_i) + \alpha \ln(y_i - p_j).$$

Why does the first imply no income effect on the discrete choice while the second generally does?

Answer. In the first specification,

$$v_{ij} = u^*(x_j, z_i) + \alpha(y_i - p_j) = [u^*(x_j, z_i) - \alpha p_j] + \alpha y_i.$$

The term αy_i is common across all choices j . Since discrete choice depends only on utility differences across alternatives, the common term drops out. That means income y_i does not affect which option is chosen, even though it appears in utility.

In the second specification,

$$v_{ij} = u^*(x_j, z_i) + \alpha \ln(y_i - p_j),$$

there is no additive term common to all alternatives that can be removed. The effect of price now depends on income because the marginal utility of income is nonlinear. A richer consumer loses a smaller fraction of utility from paying an extra dollar, so price sensitivity can vary systematically with y_i .

This distinction matters empirically. If the researcher does not observe income, the first specification is convenient because income differences do not affect choice probabilities. The second is more realistic in many settings, but it requires either observed income or additional structure to deal with the induced heterogeneity in price sensitivity.

4 Discrete Choice Basics: Horizontal and Vertical Differentiation

Question 11. Write the discrete-choice problem with an outside option.

Suppose a consumer chooses at most one inside good from $j = 1, \dots, J$ and can otherwise spend all income on an outside good 0. Explain how the budget constraint reduces to a choice among choice-specific indirect utilities.

Answer. In the discrete-choice setup, inside goods are indivisible and mutually exclusive: for $j > 0$, each $q_j \in \{0, 1\}$ and the consumer can choose at most one inside good. The outside

good q_0 absorbs remaining income. If product j is chosen, then the budget constraint becomes

$$q_0 + p_j q_j = y_i,$$

so with $q_j = 1$ we have $q_0 = y_i - p_j$. If no inside good is chosen, then $q_0 = y_i$.

This means the original utility-maximization problem can be rewritten as a comparison across alternatives:

$$V(p, y_i, z_i) = \max_{j=0,1,\dots,J} v_j(p_j, y_i, z_i),$$

where

$$v_j(p_j, y_i, z_i) = \begin{cases} u(y_i - p_j, 0, \dots, 1, \dots, 0, z_i), & j > 0, \\ u(y_i, 0, \dots, 0, z_i), & j = 0. \end{cases}$$

The outside option therefore provides the natural benchmark against which all inside goods are evaluated. Its presence also makes market shares interpretable because the demand for an inside product is the probability that its utility exceeds both the other inside goods and the outside good.

Question 12. Horizontal differentiation: solve a Hotelling-style demand problem.

Two stores are located at $x_1 = 0.25$ and $x_2 = 0.75$ on the unit interval. Consumers are uniformly distributed on $[0, 1]$, transportation cost is $t = 2$, utility is

$$v_{ij} = s - t|z_i - x_j| - p_j,$$

and prices are $p_1 = 2$ and $p_2 = 2.4$. Assume s is high enough that everyone buys from one of the two stores. Find the indifferent consumer and the aggregate demand for each store when the population is $N = 1000$.

Answer. Consumer i is indifferent between stores 1 and 2 when

$$s - 2|z_i - 0.25| - 2 = s - 2|z_i - 0.75| - 2.4.$$

Using the standard formula from the notes, the cutoff is

$$\bar{z}_1(p_1, p_2) = \frac{p_2 - p_1}{2t} + \frac{x_1 + x_2}{2}.$$

Substituting the values gives

$$\bar{z}_1 = \frac{2.4 - 2}{4} + \frac{0.25 + 0.75}{2} = 0.1 + 0.5 = 0.6.$$

So consumers with $z_i \leq 0.6$ buy from store 1, and consumers with $z_i > 0.6$ buy from store 2. Because consumers are uniformly distributed on $[0, 1]$, store 1 gets share 0.6 and store 2 gets

share 0.4. With population $N = 1000$,

$$\sigma_1 = 1000 \times 0.6 = 600, \quad \sigma_2 = 1000 \times 0.4 = 400.$$

The economic logic is simple: store 2 is farther to the right and also charges a higher price, so the indifferent consumer shifts rightward, leaving a larger market for store 1.

Question 13. Vertical differentiation: compute thresholds and market shares.

There are two goods with qualities $x_1 = 1$ and $x_2 = 2$, prices $p_1 = 1$ and $p_2 = 2.4$, and utility

$$v_{ij} = z_i x_j - p_j.$$

Consumers buy the option that gives the highest utility among good 1, good 2, and the outside good with utility 0. Suppose $z_i \sim U[0, 3]$. Find the thresholds and the market shares.

Answer. First compute the outside-option threshold,

$$\Delta_0 = \min \left\{ \frac{p_1}{x_1}, \frac{p_2}{x_2} \right\} = \min\{1, 1.2\} = 1.$$

So consumers with $z_i \leq 1$ prefer buying nothing.

Next compute the threshold between goods 1 and 2:

$$\Delta_1 = \frac{p_2 - p_1}{x_2 - x_1} = \frac{2.4 - 1}{2 - 1} = 1.4.$$

Thus:

- outside good if $z_i \in [0, 1]$,
- good 1 if $z_i \in [1, 1.4]$,
- good 2 if $z_i \in [1.4, 3]$.

Because z_i is uniform on $[0, 3]$, the shares are just interval lengths divided by 3:

$$s_0 = \frac{1}{3}, \quad s_1 = \frac{1.4 - 1}{3} = \frac{0.4}{3} \approx 0.133, \quad s_2 = \frac{3 - 1.4}{3} = \frac{1.6}{3} \approx 0.533.$$

The remaining consumers buy nothing because their willingness to pay for quality is too low. Consumers with the highest z_i values choose the high-quality product because the incremental quality is worth the higher price.

5 Multinomial Logit: Probability, Estimation, and IIA

Question 14. Derive the multinomial logit choice probability.

Assume consumer i chooses the alternative that maximizes $v_{ij} + \varepsilon_{ij}$, where the shocks are i.i.d. type-I extreme value. Show that

$$\sigma_{ij} = \frac{\exp(v_{ij})}{\sum_{k=0}^J \exp(v_{ik})}.$$

Answer. The probability that consumer i chooses product j is

$$\mathbb{P}\{j \text{ chosen}\} = \mathbb{P}\{v_{ij} + \varepsilon_{ij} \geq v_{ik} + \varepsilon_{ik} \forall k \neq j\}.$$

Condition on ε_{ij} . Then product j is chosen if

$$\varepsilon_{ik} \leq \varepsilon_{ij} + v_{ij} - v_{ik} \quad \forall k \neq j.$$

By independence,

$$\sigma_{ij} = \int \prod_{k \neq j} F(\varepsilon + v_{ij} - v_{ik}) f(\varepsilon) d\varepsilon,$$

where for type-I extreme value,

$$F(\varepsilon) = \exp\{-e^{-\varepsilon}\}, \quad f(\varepsilon) = e^{-\varepsilon} \exp\{-e^{-\varepsilon}\}.$$

Substitute these forms into the integral:

$$\sigma_{ij} = \int \exp\left\{-\sum_{k \neq j} e^{-\varepsilon - v_{ij} + v_{ik}}\right\} e^{-\varepsilon} \exp\{-e^{-\varepsilon}\} d\varepsilon.$$

Collect terms:

$$\sigma_{ij} = \int e^{-\varepsilon} \exp\left\{-e^{-\varepsilon} \sum_{k=0}^J e^{v_{ik} - v_{ij}}\right\} d\varepsilon.$$

Using the change of variable $t = e^{-\varepsilon} \sum_k e^{v_{ik} - v_{ij}}$, the integral evaluates to

$$\sigma_{ij} = \frac{1}{\sum_k e^{v_{ik} - v_{ij}}} = \frac{e^{v_{ij}}}{\sum_{k=0}^J e^{v_{ik}}}.$$

That is the multinomial logit formula.

Question 15. How do observed prices and characteristics enter the logit probability?

Suppose

$$v_{ij} = \beta'_i x_j - \alpha_i p_j,$$

and the outside option is normalized so that its characteristics vector is zero. Write the resulting choice probability and interpret the roles of β_i and α_i .

Answer. With outside option 0 normalized to have utility index 0, the logit choice probability becomes

$$\sigma_{ij} = \frac{\exp(\beta'_i x_j - \alpha_i p_j)}{1 + \sum_{k=1}^J \exp(\beta'_i x_k - \alpha_i p_k)}.$$

The parameter vector β_i measures the consumer's taste for observed product characteristics. A larger coefficient on a characteristic means that products with more of that characteristic get higher mean utility and therefore higher choice probability.

The coefficient α_i is the consumer-specific marginal disutility of price. Larger α_i means the consumer is more price sensitive. If α_i varies across consumers, then identical price changes can have different effects across market segments, which is exactly the intuition behind random-coefficient demand models.

Question 16. What is the inclusive value, and why is it useful?

Show the formula for the expected maximum utility under multinomial logit and explain its economic interpretation.

Answer. Under multinomial logit,

$$\mathbb{E} \left[\max_{j=0, \dots, J} \{v_{ij} + \varepsilon_{ij}\} \right] = \ln \left(\sum_{j=0}^J e^{v_{ij}} \right) + C,$$

where C is a constant that depends only on the extreme-value distribution.

This quantity is called the *inclusive value*. It is the consumer's ex ante expected utility from the whole choice set before the idiosyncratic taste shocks are realized. It rises when products become better, cheaper, or more numerous.

The inclusive value is important for two reasons. First, it summarizes the attractiveness of a nest or entire choice set. Second, when divided by the marginal utility of income, differences in inclusive values can be translated into money-metric welfare changes. That is why logit-style models are widely used for counterfactual consumer-surplus calculations.

Question 17. How is the multinomial logit model estimated by maximum likelihood?

Suppose micro data are available on individual choices. Write the log-likelihood and explain the intuition behind maximum likelihood estimation in this setting.

Answer. Let $q_{ij} = 1$ if consumer i chooses product j and $q_{ij} = 0$ otherwise. Given model-implied choice probabilities $\sigma_{ij}(\theta)$, the probability of observing consumer i 's realized choice vector is

$$\prod_{j=0}^J \sigma_{ij}(\theta)^{q_{ij}}.$$

Taking logs and summing over consumers yields the log-likelihood,

$$\ell(\theta) = \sum_{i=1}^N \sum_{j=0}^J q_{ij} \ln \sigma_{ij}(\theta).$$

The maximum-likelihood estimator chooses $\hat{\theta}$ to maximize this expression.

The intuition is straightforward: parameters that assign high probability to the choices that were actually observed produce a larger likelihood. Because the logit probabilities are explicit functions of the parameters, likelihood-based estimation is easy to implement and statistically efficient under correct specification.

In practice, however, MLE only delivers causal demand parameters if the regressors entering utility are exogenous or are properly instrumented or controlled for. If prices respond to unobserved product quality, naive MLE inherits the same endogeneity problem as ordinary least squares.

Question 18. How can market-segment shares be used to estimate logit demand?

Show why the multinomial logit model implies

$$\ln s_j(y, z) - \ln s_0(y, z) = v(p_j, x_j, y, z),$$

and explain when this becomes an ordinary linear regression.

Answer. For a market segment with characteristics (y, z) , logit implies

$$\sigma_j = \frac{e^{v_j}}{\sum_{k=0}^J e^{v_k}}.$$

If the outside option is normalized so that $v_0 = 0$, then

$$\sigma_0 = \frac{1}{\sum_{k=0}^J e^{v_k}}.$$

Taking the ratio gives

$$\frac{\sigma_j}{\sigma_0} = e^{v_j},$$

and therefore

$$\ln \sigma_j - \ln \sigma_0 = v_j.$$

Replacing model shares with observed market-segment shares $s_j(y, z)$ yields the estimating equation

$$\ln s_j(y, z) - \ln s_0(y, z) = v(p_j, x_j, y, z).$$

If v is linear in parameters, for example

$$v(p_j, x_j, y, z) = \beta' x_j - \alpha p_j,$$

then the left-hand side is the dependent variable in a linear regression. In that case the logit model can be estimated by least squares or instrumental variables. This is attractive because it turns a nonlinear probability model into a linearized share equation, but the usual caution about price endogeneity still applies.

Question 19. Explain the IIA property and solve the blue-bus/red-bus problem.

Originally, commuters split 50-50 between bus and train. A new bus service is introduced that is identical to the old bus service except for color. What does the multinomial logit model predict, and why is this viewed as a problem?

Answer. The IIA property says that the odds ratio between any two alternatives depends only on those two alternatives:

$$\frac{\sigma_j}{\sigma_k} = \exp(v_j - v_k).$$

This means that introducing or removing a third option changes all choice probabilities proportionally, regardless of how similar that third option is to the existing ones.

In the blue-bus/red-bus example, bus and train initially each have share 1/2. When an identical red bus is added, symmetry implies the blue bus and red bus must have the same share. IIA implies the share ratio between bus and train remains the same as before. The only way to satisfy both statements is

$$s_{\text{blue}} = s_{\text{red}} = s_{\text{train}} = \frac{1}{3}.$$

This is counterintuitive because the red bus is a clone of the blue bus, not a close substitute for the train. A realistic model should keep the total bus share near one-half and mostly split that half between the two bus services. Logit fails because it does not allow preference shocks for similar products to be correlated.

Question 20. Why do multinomial-logit price elasticities often look unrealistic?

Suppose utility is $v_j = \beta'x_j - \alpha p_j$. Explain why logit implies restrictive substitution patterns, and interpret the example in which Coke has share 1/2, Pepsi has share 1/6, and coffee has share 1/3.

Answer. Under simple logit, the own- and cross-price elasticities have the form

$$e_{jj} = -\alpha p_j(1 - \sigma_j), \quad e_{jk} = \alpha p_k \sigma_k \quad (j \neq k).$$

The key problem is that the cross-elasticity with respect to product k depends only on product k 's share, not on how similar k is to j . So substitution is governed by market shares, not by product proximity in characteristics space.

In the example, suppose Coke has share 1/2, Pepsi has share 1/6, and coffee has share 1/3. If Coke's price rises, the simple logit model predicts that consumers leaving Coke substitute twice as much toward coffee as toward Pepsi, because coffee's share is twice Pepsi's share:

$$\frac{e_{\text{Coke, coffee}}}{e_{\text{Coke, Pepsi}}} = \frac{1/3}{1/6} = 2.$$

That is implausible. Pepsi is much closer to Coke than coffee is, so a realistic model should produce more diversion toward Pepsi.

This is one of the main reasons why empirical IO moved from simple logit toward mixed logit, nested logit, and richer random-coefficient models.

6 Mixed Logit and Simulation

Question 21. How does mixed logit relax IIA?

Suppose utility is

$$v_{ij} = \beta'_i x_j - \alpha_i p_j,$$

where (β_i, α_i) vary across consumers. Explain why this model no longer has IIA at the market level.

Answer. Conditional on a consumer's type (β_i, α_i) , the choice probabilities are still logit:

$$\sigma_{ij}(\beta_i, \alpha_i) = \frac{\exp(\beta'_i x_j - \alpha_i p_j)}{\sum_{k=0}^J \exp(\beta'_i x_k - \alpha_i p_k)}.$$

But the econometrician does not observe (β_i, α_i) , so the market-level choice probability

integrates over the distribution of consumer types:

$$\sigma_j = \int \sigma_{ij}(\beta, \alpha) f(\beta, \alpha \mid y_i, z_i) d\beta d\alpha.$$

Because the market share is an average of many consumer-specific logits, the ratio σ_j/σ_k is no longer a simple function of only j and k . It depends on all products through the integral and through the way different consumer types sort across alternatives.

Economically, this means consumers who especially like carbonated drinks, SUVs, or luxury cars tend to substitute toward similar products when their favorite option becomes more expensive or unavailable. The correlation in substitution comes from heterogeneity in tastes rather than from directly correlated extreme-value shocks.

Question 22. Explain the difference between observed and unobserved consumer heterogeneity.

Use the beverage example from the chapter: consumers may value carbonated drinks differently depending on whether they are teenagers and depending on an unobserved taste shock.

Answer. Observed heterogeneity means the econometrician sees the characteristic that changes tastes. For example, if

$$\beta_i = 0.1 + 0.2 \cdot \text{teenager}_i,$$

then teenagers value carbonated drinks more than non-teenagers, and the researcher can model that directly because teenage status is observed.

Unobserved heterogeneity adds a component that the econometrician does not observe, such as

$$\beta_i = 0.1 + 0.2 \cdot \text{teenager}_i + \nu_i.$$

Now even among teenagers, some consumers intrinsically love carbonated drinks more than others. This unobserved variation matters because it changes substitution patterns: consumers with high ν_i are more likely to replace Coke with Pepsi rather than with coffee.

Observed heterogeneity improves realism, but by itself it only removes IIA after aggregating across observed segments. Unobserved heterogeneity is what removes IIA even within a market segment defined by observables.

Question 23. How are mixed-logit probabilities estimated in practice?

Describe simulated maximum likelihood for mixed logit. Why is simulation needed?

Answer. Mixed logit choice probabilities are integrals over the distribution of random

coefficients:

$$\sigma_j = \int \frac{\exp(v_{ij}(\beta, \alpha))}{\sum_{k=0}^J \exp(v_{ik}(\beta, \alpha))} f(\beta, \alpha \mid y_i, z_i) d\beta d\alpha.$$

These integrals are usually not available in closed form, so simulation is used.

In simulated maximum likelihood, the researcher:

1. draws R values (β^r, α^r) from the assumed distribution of consumer types,
2. computes the logit probability for each draw,
3. averages across draws to approximate the mixed-logit probability,

so that

$$\hat{\sigma}_j = \frac{1}{R} \sum_{r=1}^R \frac{\exp(v_{ij}(\beta^r, \alpha^r))}{\sum_{k=0}^J \exp(v_{ik}(\beta^r, \alpha^r))}.$$

These simulated probabilities are inserted into the log-likelihood and the parameters are chosen to maximize it.

Simulation is needed because random coefficients create rich substitution patterns precisely by integrating over many latent consumer types. That flexibility comes at the cost of numerical integration.

Question 24. What is the role of quadrature in mixed-logit estimation?

Explain the idea of Gauss-Hermite quadrature and when it may be preferable to plain Monte Carlo simulation.

Answer. Gauss-Hermite quadrature is a deterministic numerical integration method designed for integrals of the form

$$\int_{-\infty}^{\infty} f(z) e^{-z^2} dz.$$

Because normal random coefficients can be re-expressed in that form after a change of variables, quadrature can be used to approximate expectations under normal distributions.

The basic idea is to replace the integral with a weighted sum at carefully chosen nodes:

$$\mathbb{E}[f(Y)] \approx \pi^{-1/2} \sum_{r=1}^R \omega_r f(\mu + \sqrt{2}\sigma z_r),$$

where z_r are quadrature nodes and ω_r are weights.

Compared with plain Monte Carlo, quadrature can be more accurate when the number of random coefficients is small because it uses structured evaluation points rather than random draws. But quadrature suffers from the curse of dimensionality as the number of random coefficients grows. Monte Carlo methods scale more easily in high dimension, which is why they are common in empirical IO.

7 Nested Logit

Question 25. Write the nested-logit choice probability and interpret its two pieces.

Suppose products are grouped into nests $g = 1, \dots, G$. Write the nested-logit choice probability for product j in nest g , define the inclusive value I_g , and explain the decomposition into within-nest and across-nest choice.

Answer. A standard nested-logit probability can be written as

$$\sigma_j = \underbrace{\frac{\exp(v_j/\lambda_g)}{\sum_{k \in g} \exp(v_k/\lambda_g)}}_{\text{within-nest choice}} \times \underbrace{\frac{\exp(\lambda_g I_g)}{\sum_{h=0}^G \exp(\lambda_h I_h)}}_{\text{nest choice}},$$

where

$$I_g = \ln \left(\sum_{k \in g} \exp(v_k/\lambda_g) \right).$$

The first term is the probability of choosing product j conditional on already being in nest g . The second term is the probability of choosing nest g itself. This decomposition is attractive because it introduces a simple form of correlation across similar products: alternatives in the same nest share a common nest-level taste component.

Economically, nested logit says substitution is stronger within a category than across categories. If one compact sedan becomes unavailable, demand is more likely to flow to other compact sedans than to pickup trucks. The dissimilarity parameter λ_g controls how tightly linked products are within the nest; in the common parameterization, smaller values imply stronger within-nest correlation.

Question 26. How does the Berry nested-logit estimating equation work?

Explain the equation

$$\ln s_j - \ln s_0 = v_j + (1 - \lambda_g) \ln s_{j|g},$$

and why both price and within-group share are endogenous.

Answer. Nested logit implies that the log share relative to the outside good can be written as the sum of two pieces: the mean utility of the product itself and a correction term capturing how attractive its nest is. That produces

$$\ln s_j - \ln s_0 = v_j + (1 - \lambda_g) \ln s_{j|g},$$

where $s_{j|g}$ is the share of product j within its own nest.

This equation is useful because it looks like a regression. If v_j is linear in parameters, the only nonlinear object is the within-group-share term. But two regressors are endogenous.

First, price is endogenous for the usual reason: firms raise prices when they observe favorable

unobserved quality shocks. Second, the within-group share is endogenous because it mechanically depends on the unobserved attractiveness of product j and its nest. A product with positive unobserved quality gets both a larger overall share and a larger within-nest share. That is why estimation requires instruments for both price and $\ln s_{j|g}$. Typical instruments include cost shifters, the number of products in a nest, or characteristics of other products that shift competitive pressure without directly entering utility.

8 Unobserved Product Quality, Price Endogeneity, and Inversion

Question 27. How does unobserved product quality create endogeneity in logit demand?

Suppose utility in market t is

$$u_{ijt} = \beta' x_{jt} - \alpha p_{jt} + \xi_{jt} + \varepsilon_{ijt},$$

where ξ_{jt} is observed by consumers and firms but not by the econometrician. Show how the logit model can be inverted for ξ_{jt} and explain the role of instruments.

Answer. Under simple logit, the market share of product j satisfies

$$s_{jt} = \frac{\exp(\beta' x_{jt} - \alpha p_{jt} + \xi_{jt})}{1 + \sum_{k=1}^J \exp(\beta' x_{kt} - \alpha p_{kt} + \xi_{kt})}.$$

The outside-good share is

$$s_{0t} = \frac{1}{1 + \sum_{k=1}^J \exp(\beta' x_{kt} - \alpha p_{kt} + \xi_{kt})}.$$

Taking the ratio gives

$$\ln s_{jt} - \ln s_{0t} = \beta' x_{jt} - \alpha p_{jt} + \xi_{jt}.$$

Hence the unobserved product-market quality can be written as

$$\xi_{jt} = \ln s_{jt} - \ln s_{0t} - (\beta' x_{jt} - \alpha p_{jt}).$$

This inversion is extremely useful: it turns an unobserved demand shock into a residual that depends on observed shares and parameters. The remaining identification problem is that p_{jt} is usually correlated with ξ_{jt} , because firms set higher prices when they know a product has favorable unobserved quality.

Therefore estimation uses instruments w_{jt} satisfying

$$\mathbb{E}[\xi_{jt} \mid w_{jt}] = 0.$$

Typical instruments are cost shifters or competitive-environment variables that affect prices but not consumer demand directly.

Question 28. What happens if unobserved product quality is fixed across markets?

Suppose $\xi_{jt} = \xi_j$, constant over time or across markets for each product. Explain how this simplifies estimation.

Answer. If unobserved quality is product-specific but market-invariant, then the logit inversion becomes

$$\ln s_{jt} - \ln s_{0t} = \beta' x_{jt} - \alpha p_{jt} + \xi_j.$$

This is just a linear panel regression with product fixed effects. The fixed effects absorb persistent brand value or other time-invariant unobserved product characteristics.

That is a major simplification. Instead of solving a high-dimensional nonlinear inversion problem, the researcher can difference out or partial out the product fixed effects using standard panel-data methods. The resulting estimates often serve as a useful starting point for richer random-coefficient specifications.

The limitation is that many important demand shocks are not fixed across markets. Promotions, local advertising, local shelf placement, and time-varying reputation shocks all create market-specific unobservables, which is why the more general BLP framework remains necessary in many applications.

Question 29. Explain the BLP contraction mapping.

In the random-coefficients logit model, define the average utility

$$\delta_{jt} = \beta_0' x_{jt} - \alpha_0 p_{jt} + \xi_{jt}.$$

Explain why δ_t can be solved from observed market shares using the contraction mapping

$$T_t(\delta_t) = \delta_t + \ln s_t - \ln \sigma_t(\delta_t, \theta_2).$$

Answer. In random-coefficients demand, market shares are nonlinear functions of both the average utility vector δ_t and the nonlinear taste-heterogeneity parameters θ_2 . For a given value of θ_2 , the model predicts shares

$$\sigma_t(\delta_t, \theta_2).$$

The data provide actual shares s_t . So the econometric problem is to find the vector δ_t that solves

$$s_t = \sigma_t(\delta_t, \theta_2).$$

BLP show that for fixed θ_2 , the operator

$$T_t(\delta_t) = \delta_t + \ln s_t - \ln \sigma_t(\delta_t, \theta_2)$$

forms a contraction mapping. Intuitively, if the model underpredicts the share of a product, the update raises that product's average utility; if the model overpredicts its share, the update lowers average utility. Repeating this update pushes the predicted shares toward the observed ones.

Because the mapping is a contraction, three desirable properties follow: there is a unique fixed point, iterating from any starting value converges to that fixed point, and the fixed point is exactly the average-utility vector that rationalizes the observed shares for the chosen θ_2 .

Once $\delta_t(\theta_2)$ is recovered, the unobserved quality is just

$$\xi_{jt}(\theta) = \delta_{jt}(\theta_2) - \beta_0' x_{jt} + \alpha_0 p_{jt}.$$

That converts an otherwise intractable nonlinear demand system into something estimable by GMM.

Question 30. Why are some parameters called “linear” and others “nonlinear” in BLP?

Explain the distinction between θ_1 and θ_2 , and show how the linear parameters can be recovered once $\delta(\theta_2)$ is known.

Answer. The random-coefficients BLP model separates parameters into two blocks.

The *nonlinear parameters*, denoted θ_2 , govern taste heterogeneity: the variances and covariances of random coefficients and any interactions with observed demographics. They are called nonlinear because the market-share equations depend on them through simulation or numerical integration, and there is no closed-form linear regression that solves for them.

The *linear parameters*, denoted θ_1 , are the coefficients that enter average utility linearly, such as mean tastes for observed characteristics and the mean price coefficient. Once $\delta(\theta_2)$ has been recovered from the contraction mapping, these parameters solve the linear relationship

$$\delta = X_1 \theta_1 + \xi,$$

where X_1 collects observed characteristics and price.

Given instruments W , the optimal linear parameters for a fixed θ_2 are obtained by IV/GMM regression:

$$\hat{\theta}_1(\theta_2) = (X_1' W \Phi^{-1} W' X_1)^{-1} X_1' W \Phi^{-1} W' \delta(\theta_2).$$

This is why the computational search is usually conducted only over θ_2 . For each candidate θ_2 , the contraction mapping gives δ , the IV step gives θ_1 , and the implied residuals ξ are used to evaluate the GMM objective.

Question 31. What is the BLP GMM objective, and what are the main instrument choices?

State the moment condition and summarize the logic of the main instrument families used in differentiated-products demand estimation.

Answer. Once the mean utilities have been inverted, the unobserved quality term is

$$\xi(\theta) = \delta(\theta_2) - X_1\theta_1.$$

Identification comes from moment conditions of the form

$$\mathbb{E}[\xi_{jt}(\theta) w_{jt}] = 0,$$

where w_{jt} includes exogenous product characteristics and excluded demand instruments. Stacking the moments across products and markets yields the GMM objective

$$Q(\theta) = \xi(\theta)' W \Phi^{-1} W' \xi(\theta).$$

The estimator chooses the parameter vector that makes the residual quality term as orthogonal as possible to the instruments.

The main instrument families are:

- **Cost shifters.** These are the cleanest instruments when available. They move prices through costs but do not directly affect utility.
- **Hausman-type instruments.** Prices of the same product in other markets can be informative if cost shocks are correlated across markets while demand shocks are market-specific.
- **BLP-type instruments.** Characteristics of rival products, or sums of characteristics of products owned by the same firm and by rival firms, proxy for competitive pressure and markup incentives.
- **Differentiation instruments.** Functions of the distances between products in characteristics space capture the idea that nearby substitutes affect pricing more strongly than distant products.

Each family targets the same identification problem: prices are endogenous because firms observe product quality that the econometrician does not. Good instruments change equilibrium prices without directly moving the unobserved taste component.

Question 32. When is a control-function approach useful, and how does it work?

Explain the control-function logic for discrete-choice demand estimation when the BLP share-inversion approach is inconvenient, for example because consumers face different choice sets.

Answer. The BLP inversion is most natural when the researcher observes common market-level shares and consumers in the same market face the same choice set. When choice sets vary at the individual level, or when shares are not the convenient sufficient statistic, a control-function approach can be more practical.

The idea is to model the endogenous price-setting equation directly:

$$p_t = P(x_t, w_t, \xi_t),$$

where x_t are demand-side observables, w_t are excluded variables that affect price but not utility directly, and ξ_t is the unobserved product quality. If the first-stage pricing equation can be estimated and inverted, the researcher recovers a proxy for the endogenous component of price. That proxy is then included in the utility specification as a control function.

A simple example decomposes the unobserved quality as

$$\xi_{jt} = \rho\nu_{jt} + \eta_{jt},$$

where ν_{jt} is the anticipated component that affects price and η_{jt} is an innovation orthogonal to the instruments. Regressing price on exogenous observables and cost shifters gives an estimate of ν_{jt} , which is then inserted into the demand equation. Conditional on that proxy, the remaining error is orthogonal to the instruments.

So the control-function strategy solves the same endogeneity problem as IV-BLP, but it does so by modeling the price equation explicitly rather than by recovering mean utility from market shares.

A Compact Formula Sheet

Core formulas from Chapter 4

Duality and demand

$$e(u, p) = \min_q p \cdot q \text{ s.t. } u(q) \geq u, \quad v(p, x) = e^{-1}(p, x),$$

$$h_i(u, p) = \frac{\partial e(u, p)}{\partial p_i}, \quad d_i(p, x) = h_i(v(p, x), p),$$

$$d_i(p, x) = -\frac{\partial v(p, x)/\partial p_i}{\partial v(p, x)/\partial x} \quad (\text{Roy's identity}).$$

Expenditure share equation

$$w_i = \frac{p_i q_i}{x} = \frac{\partial \ln e(u, p)}{\partial \ln p_i}.$$

AIDS share equation

$$w_i = \alpha_i + \sum_j \gamma_{ij} \ln p_j + \beta_i \ln \left(\frac{x}{P} \right).$$

Multinomial logit

$$\sigma_{ij} = \frac{\exp(v_{ij})}{\sum_{k=0}^J \exp(v_{ik})}, \quad \mathbb{E} \max_j (v_{ij} + \varepsilon_{ij}) = \ln \left(\sum_j e^{v_{ij}} \right) + C.$$

Logit share inversion

$$\ln s_{jt} - \ln s_{0t} = \beta' x_{jt} - \alpha p_{jt} + \xi_{jt}.$$

Mixed logit

$$\sigma_j = \int \frac{\exp(v_{ij}(\beta, \alpha))}{\sum_{k=0}^J \exp(v_{ik}(\beta, \alpha))} f(\beta, \alpha) d\beta d\alpha.$$

Nested logit

$$\sigma_j = \frac{\exp(v_j/\lambda_g)}{\sum_{k \in g} \exp(v_k/\lambda_g)} \times \frac{\exp(\lambda_g I_g)}{\sum_h \exp(\lambda_h I_h)}, \quad I_g = \ln \sum_{k \in g} \exp(v_k/\lambda_g).$$

BLP contraction mapping

$$T_t(\delta_t) = \delta_t + \ln s_t - \ln \sigma_t(\delta_t, \theta_2).$$